Sourcing list

DIY CNC epoxy-granite mill · October 2026

Parts to buy for the machine in `model/DIYCNC.FCStd` / `docs/report.txt`, sourced on Amazon (US) where it makes sense. Prices and stock were checked on **2026-10-09**. "?" means the price was not visible (Amazon blocks automated page reads); check the live listing. Shipping and tax are not included. Off-Amazon picks are marked "not Amazon": they are there because Amazon doesn't carry the part or is much dearer. Rows marked (opt) are optional and are left out of the totals.

The same list is in `docs/sourcing.csv` for a spreadsheet.

Totals

Section	Priced	Unpriced items
Motion and drives	$522.50	0
Spindle and electronics	$367.78	14
Castings and frame	$1,267.80	0
Enclosure, coolant and covers	$573.92	3
Total	**$2,732.00**	**17**

The unpriced items are mostly the 120 V VFD, the electrical parts, the ACM sheet, the sump pump and the spiral cover. My rough estimate for them is another $450-750 (not checked), so the build comes to roughly $3.2-3.5k before shipping.

Where the money goes: the frame section: casting materials ~$360, aluminium plate ~$240, steel tube and plate ~$245, rods and fasteners ~$190, risers ~$110. Then the enclosure and coolant ($574), the spindle and VFD, and the closed-loop motors ($162).

Cheapest places to cut: buy the tube, top-plate steel and risers from a local steel yard (offcuts or full lengths you cut yourself; estimated ~$100-150 less than the listings here) and the silica flour from a pottery supplier (~$40 less); skip the flood coolant and enclosure at first (~$550, see the README cost section); AliExpress versions of the rails, ballscrews, spindle and VFD (often 30-50 % less, slower shipping). Retail cut-to-size shops (e.g. Metal Supermarkets) were checked and cost more than these listings, about $820 for the tube, risers and aluminium.

Motion and drives

Item	Spec	Qty	Listing	Price	Line	Notes
Y rails + blocks	HGR15 x2 at 600 mm + 4x HGH15CA	1 kit	iMetrx HGR15 kit, 2x 600 mm + 4x HGH15CA	$55.99	$55.99	Used at full length, no cutting. The listing calls the blocks "aluminum", which is probably wrong: check them for play.
Z rails + blocks	HGR15 x2 at 300 mm + 4x HGH15CA	1 kit	BEVDICNC 2x HGR15-300 + 4x HGH15CA	$35.69	$35.69	Alt: CNCMANS B09N73DY32 $34.98
Y ballscrew kit	SFU1605-550, BK12/BF12, DSG16H	1	Ballscrew kit DSG16H + SFU1605-550 + BK/BF12	$39.39	$39.39	Ships with a 6.35x10 coupler: use the 8x10 below
Z ballscrew kit	SFU1204-300, BK10/BF10, DSG12H	1	CNCYEAH SFU1204 300 mm kit	$32.69	$32.69	Ships with a 6.35x8 coupler: use the 8x8 below
Z motor bracket	NEMA 23 + BK10 (HM10-57)	1	ANWOK NEMA 23 steel bracket, SFU1204	$32.02	$32.02	Same bracket as your X one
Y motor bracket	NEMA 23 + BK12 (HM12-57)	1	ANWOK NEMA 23 steel bracket, SFU1605	$31.36	$31.36	Alt: CNCMANS HM12-57 B09QS2S3BT $28.98
X/Y closed-loop kits	23HS30-5004D-E1000 2.0 Nm + CL57T	2	STEPPERONLINE closed-loop kit 2.0 Nm + CL57T	$81.00	$162.00	Exact model. Low stock when checked.
Z driver (owned motor)	DM542T	1	STEPPERONLINE DM542T	$32.99	$32.99	Check that your motor is 4.2 A or less
Motor power supply	48 V, 3 drivers	1	Mean Well LRS-350-48 (7.3 A)	$37.13	$37.13	Trim to about 46 V: CL57T max input is 48 V
Couplers, Y	8x10 mm, D25 L30	1 (3-pack)	CNCTOPBAOS 8x10 plum coupling, 3 pcs	$13.99	$13.99	Spares; check that D25 fits inside the bracket
Couplers, Z	8x8 mm, D25 L30	1 (2-pack)	Yeebyee 8x8 plum coupling, 2 pcs	$12.89	$12.89	The spare covers X if your kit came with 6.35x8
Grease gun	Mini, 3 oz cartridges	1	Performance Tool W54205 mini grease gun kit	$17.59	$17.59	
Needle tip	For the blocks' M4 nipples	1	LubriMatic 05-037 needle	$6.98	$6.98	
Grease	Lithium NLGI 2	1 (3x3 oz)	LubriMatic 11312, 3x 3 oz	$11.79	$11.79	
(opt) M4 grease nipples	M4x0.7 straight	1 pack	HARFINGTON 15 pcs M4x0.7	$13.94	$13.94	Only if the block nipples are missing or the coupler won't seat

Priced subtotal: $522.50

Spindle and electronics

Item	Spec	Qty	Listing	Price	Line	Notes
Spindle kit	2.2 kW 80 mm ER20 water-cooled + 220 V VFD + clamp	1	RATTMMOTOR 2.2 kW ER20 kit with VFD, clamp, pump	$250.96	$250.96	Its VFD needs 220 V in: keep it for the 240 V upgrade. Its pump is 220 V too. Alt with 13 ER20 collets included: B07JB3TJKR (price not seen).
VFD for 120 V now	110-120 V in, 220 V 3-phase out, 400 Hz	1	Huanyang 1.5 kW step-up VFD (110 V in, 220 V out)	?	?	Caps the spindle at ~1.5 kW until 240 V. Ask the seller to confirm 400 Hz. Alt: a 110 V-out HY02D211B (B0775F4G47) with the V/f curve set to 110 V at 200 Hz gives full torque up to 12k rpm.
ER20 collet set	1-13 mm	1	Chiloskit ER20 13 pcs	$25.21	$25.21	Skip if the kit includes a set
Spindle water pump	110 V submersible, ~80 W	1	110 V 80 W submersible pump for water-cooled spindles	?	?	Make sure it is the 110 V version; many are 220 V. Runs in a 10-20 L bucket.
Spindle coolant	Distilled water + 10-20 % RV antifreeze	1	Local store	?	?	Stops algae and corrosion
(opt) Radiator	240 mm with fans	1	ASHATA 240 mm radiator + 2 fans	?	?	Only if the bucket warms up on long jobs
Controller	FluidNC, 6 axes, external drivers, RS485, 0-10 V	1	Elecrow Corgi ESP32 FluidNC controller (not Amazon)	$49.00	$49.00	Runs gSender or the web UI; no Windows license. Alt: DDCSV3.1 standalone B07SR854D1 (about $250-330 elsewhere).
0-10 V spindle module	For the controller	1	Elecrow FluidNC 0-10 V module	$11.00	$11.00	Check whether the Corgi already has 0-10 V built in
Spindle pump relay	SSR, DC in, 120 V AC out	1	Aexit SSR-25DA	?	?	Runs the 110 V spindle water pump whenever the spindle runs. Fuse the branch.
Homing sensors	Inductive NPN NO, M12	1 (5-pack)	LJ12A3-4-Z/BX, 5 pcs	?	?	Power them from 12 V, not 5 V
Tool setter	NC, magnetic base	1	Z touch tool setter	~$31.61	$31.61	Price may belong to a neighbouring listing
E-stop	22 mm mushroom, latching, in a box	1	Shopcorp E-stop with box	?	?	Wire it to cut drive enable and VFD run in hardware
Mains inlet, logic	IEC C14 with switch + fuse	1 (4-pack)	Oiyagai C14 inlet, switch, fuse	?	?	Logic, 48 V supply and pump only (6 A switch)
Mains inlet, VFD	IEC C20, 16/20 A	1	Interpower 83030420	?	?	The VFD draws 15-20 A on 120 V: a 20 A circuit
Logic power supply	24 V DIN rail	1	Mean Well HDR-60-24	?	?	

Item	Spec	Qty	Listing	Price	Line	Notes
Spindle cable	14/4 shielded, 3 m	1 (10 ft)	LINKSQ 14/4 shielded	?	?	Shield to earth at the VFD end
Stepper/sensor cable	18/4 shielded	1 (50 ft)	Linear Motion Labs 18/4 shielded	?	?	
Electrical box	Steel, room for VFD + PSUs	1	VEVOR 20x16x8 in enclosure	?	?	16x12x6 (B09247D9LH) is tight with the VFD inside. Add a filtered fan.
DIN rail + terminals	UK2.5B kit	1	Erayco terminal block kit	?	?	
Cable glands	PG7-PG19	1 (50 pcs)	Bates nylon glands	?	?	

Priced subtotal: $367.78 + 14 unpriced: VFD for 120 V now, Spindle water pump, Spindle coolant, Spindle pump relay, Homing sensors, E-stop, Mains inlet, logic, Mains inlet, VFD, Logic power supply, Spindle cable, Stepper/sensor cable, Electrical box, DIN rail + terminals, Cable glands

Castings and frame

Item	Spec	Qty	Listing	Price	Line	Notes
Epoxy	~10.6 kg low-viscosity laminating epoxy	1 (3 gal)	US Composites 635 Thin, 2 gal + 1 gal (not Amazon)	$185.30	$185.30	About $15/kg. Amazon alt: TotalBoat 2:1 Slow 3 gal B07N98VC3Z $354.99. Avoid deep-pour casting resins.
Fines	Silica flour 200-325 mesh, 10-15 kg	1 (50 lb)	Florock ground silica, 50 lb	$94.37	$94.37	Mesh not stated. A pottery supplier sells 200 mesh for about half (e.g. $53 / 50 lb). Wear a respirator.
Aggregate	~106 kg graded sand + pea gravel	~5 bags	Quikrete sand + Vigoro pea gravel (Home Depot)	~$7/bag	$40.00	Buy locally. Wash and dry fully.
Mould release wax	Partall Paste #2	1	Partall Paste #2, 12 oz	$18.52	$18.52	
Mould release PVA	Partall Film #10	1 qt	Partall PVA, quart	$22.99	$22.99	
Square tube 2x2x1/8, 12 in	Base tubes (split sockets)	4	Rocaris 2x2x1/8 steel square tube, 12 in	$24.99	$99.96	Bought at length. Steel-yard remnants are cheaper (0.120 wall is fine). Plain steel, not galvanized.
Square tube 2x2x1/8, 18 in	Beam tubes	2	2TwentyTwo 2x2x1/8 steel tube, 18 in	$18.42	$36.84	Drill 4x 14 mm holes per tube (drill press). Remnants are fine: 18 in +/- 1/4 in, any surface rust brushed off.
Square tube 1-1/2x1-1/2x1/8, 12 in	Upright posts	4	1.5x1.5x1/8 mild steel tube (12)	$13.73	$54.92	

Item	Spec	Qty	Listing	Price	Line	Notes
Square tube 1-1/2x1-1/2x1/8, 6 in	Base spigots (2) + legs (4)	6	1.5x1.5x1/8 mild steel tube (6)	$5.52	$33.12	Printed sleeves take up the spigot gap in the 2 in tube; file the inside weld seam. Leg feet: an M12 nut (from the 20-pack) bonded into a printed plug in each leg end.
Steel plate 3/8 in	Upright top plates 2x 68 x 168 mm (2-11/16 x 6-5/8 in)	2 cut pieces	A36 plate cut to size (Midwest Steel or a local yard)	~$15	$15.00	Order cut to size so nothing needs cutting; add it to the riser order.
Rail risers 3/4 x 1-1/4 CR	Y: 2x 600 mm	2	Midwest Steel 1018 cold-rolled flat bar (not Amazon)	~$4.50/lb	$58.00	Not on Amazon. Cut to length by Midwest (cut fee not included).
Rail risers 5/8 x 1-1/4 CR	X: 2x 400, Z: 2x 300 mm	4	Midwest Steel 1018 cold-rolled flat bar (not Amazon)	~$4.50/lb	$56.00	Same
Aluminium plates	Cast plate: table 3/4 x 10.5 x 13 in, X 5/8 x 7 x 16.5, Z 5/8 x 6.25 x 12.25	3 cut pieces	Midwest ATP-5 / MIC-6 cast plate, cut to size (not Amazon)	~$242	$242.00	Amazon alt is about $458: 6061 3/4x12x18 B007B4XUFU $215.98 + 5/8x12x24 B0DYG69W17 $241.97 (12x12 is too small for the table).
M12 threaded rod	4x 293 + 2x 358 mm	2 (1 m)	M12-1.75 x 1 m plain steel rod	$28.96	$57.92	Cut with a hacksaw. Confirm it is one rod per order.
M12 coupling nuts	6	1 (8-pack)	Hillman 44770 M12 coupling nut	$16.97	$16.97	
M12 nuts	12	1 (20-pack)	Hillman 2468 M12 hex nut	$8.92	$8.92	
M12 washers	12	1 (25-pack)	M12 flat washer, hardened	$11.84	$11.84	
M10 x 150 cap screws	8, 12.9	2 (5-pack)	Aexit M10x150 12.9	$23.53	$47.06	M10 washers: buy locally
Dowel pins 10 x 30	4 (locating only)	1 (14-pack)	10 x 30 mm 304 stainless dowel / shelf pin, 14 pcs (Walmart)	$18.49	$18.49	They only locate the castings (the bolts and rods carry the load), so plain pins are fine. Was a $85 hardened 20-pack.
Leveling feet	M12, ~40 kg each	1 (4-pack)	Antrader M12 leg levelers D60	$23.99	$23.99	
O-ring cord	NBR 3 mm, 2.5 m	1 (3 m)	uxcell nitrile round strip 3 mm x 3 m	$6.56	$6.56	
Cord splice glue	Cyanoacrylate	1	Loctite 406, 20 g	$9.86	$9.86	
M5 cast-in inserts	29 (11 enclosure + 18 riser hold-downs)	1 (30)	uxcell knurled brass M5 x 16 mm	$11.08	$11.08	Screw each to the mould floor with a greased screw so it ends up flush.
M6 cast-in inserts	6 (Y BK12/BF12 on the base)	1 pack	Knurled brass M6 insert, ~15 mm long	~$8	$8.00	Search "M6 knurled brass insert 15mm".

Item	Spec	Qty	Listing	Price	Line	Notes
X support pads	1/4 x 2-1/2 in flat bar, ~4 in	1 offcut	Steel yard remnant	~$5	$5.00	Two pads (49 and 36 mm), drilled and tapped M6, cast flush in the beam face.
Riser hold-down screws	DIN 7984 low-head M5 x 20 (8) and M5 x 25 (10)	2 packs	Low-head socket cap screws	~$12	$12.00	Low head (3.5 mm) so they sit below the rail in the 4.5 mm counterbores.
Support screws	M6 x 40 (6, Y); M6 x 35 + 12 DIN 433 small washers (6, X)	2 packs + washers	M6 socket cap screws 12.9	~$10	$10.00	Two small washers (10 mm OD, fit the BK counterbore) under each X screw head so it stops short of the beam tube.
M5 x 35 cap screws	11	1 (25)	M5x35 12.9 SHCS	$8.99	$8.99	
Cap screw assortment	M3-M6	1	Mywish 900 pcs 12.9 kit	$21.99	$21.99	
Rail screws	M4 x 16	1 (100)	iexcell M4x16 12.9	$8.26	$8.26	
Block screws	M4 x 20	1 (50)	iexcell M4x20 12.9	$7.86	$7.86	Check length against the plate thickness
PETG filament	1 kg	1	SUNLU PETG 1 kg	$15.99	$15.99	Cable chain, socket liners, sleeves, plugs

Priced subtotal: $1,267.80

Enclosure, coolant and covers

Item	Spec	Qty	Listing	Price	Line	Notes
2020 extrusion	4x766, 2x756, 4x684, 2x600, 3x560 mm	2 (10x 1000 mm)	KathyLiely 2020 T-slot 1000 mm, 10 pcs	$49.99	$99.98	One piece per stick; 5 sticks spare. Take the 2x 244 from your 400s.
ACM sheet	3 mm, all four wall/roof panels	1 (4x8 ft)	Sign supply / Piedmont Plastics (not Amazon)	?	?	The 726x684 sides don't fit any in-stock Amazon sheet. One 4x8 covers all four panels and the drip shield. Amazon 24x36 2-pack (B0F18ZQ7DJ, $65.99) covers back + roof only.
Door	Clear polycarbonate 560x756	1	SIBE polycarbonate 1/8 in, 24x36	$24.63	$24.63	
Corner brackets	~30	1	JCSPBYL 30 sets 2020 corner brackets	$12.99	$12.99	
T-nuts	M5 drop-in, 100	1	POLISI3D 100 M5 drop-in T-nuts	$7.65	$7.65	
Button-head screws	M5 x 10, 100	1	uxcell M5x10 button head, 100	$13.99	$13.99	

Item	Spec	Qty	Listing	Price	Line	Notes
Hinges + handle	2020, 6 mm slot	1 kit	KOOTANS 2020 hinge and handle kit	$14.59	$14.59	M4 or M5 not stated
Gas struts	2x ~50 N, 200 mm	1 pair	SBOTTON 200 mm gas struts, 50 N	$38.76	$38.76	Check force against the door weight
Door catch	Magnetic	1 (4-pack)	Onarway cabinet magnets	$13.99	$13.99	
(opt) Door interlock	2NC key switch	1	CZ-93B safety switch	$22.13	$22.13	Wire into the E-stop or feed-hold loop
Panel trim	U-strip for 3 mm panels	1	BLCCLOY 2020 panel U-strip, 25x 420 mm	$17.99	$17.99	10.5 m; just short of every edge
Brush seal	For 2x 72 mm motor cutouts	1 m	SLZ dust-shoe brush strip 50 mm	$20.00	$20.00	
Bulkhead	1/4 NPT, thin panel	1 (2-pack)	FloFlex 1/4 in bulkhead, short	$9.99	$9.99	Measure the hole size first
Sump	Full-size steam-table pan, 4 in deep (~13 L)	1	Winco SPJH-104, 22 ga stainless (not Amazon)	$22.16-26.81	$26.81	Not found on Amazon; restaurant-supply stores (KaTom, GoFoodservice $23.63) or a local one. Slides in under the base.
Chip basket	1/3-size pan, 2-1/2 in deep	1	CAC STPT-S25-2 1/3 pan (not Amazon)	$9.98	$9.98	Solid pan: drill 6 mm holes in the floor (or buy a perforated 1/3 x 2-1/2 in pan). Sits at the front, under the drain.
Sump filter mesh	100 um	1	100 micron filter sock (cut open)	$10.59	$10.59	Cut open and clamp in the printed filter frame
Magnet bar	100 x 25 mm	1	ULIBERMAGNET 12000 Gs filter rod 100x25	$29.99	$29.99	Lies on the sump floor and catches fine steel
Flood coolant pump	24 V brushless submersible, ~800 L/h, 5 m head, <= 60 mm tall	1	Search Amazon: "24V brushless submersible pump 800L/H 5M"	?	?	Low enough for the 4 in sump and no mains voltage in the coolant. Runs from the 24 V supply via the controller's MOSFET output. Replaces the $113 120 V pump.
Return tubing	8 mm OD PU	1 (5.5 m)	uxcell 8 mm PU tubing	$10.19	$10.19	+$1.21 shipping
Drain nipple	1/2 NPT brass	1 (2-pack)	Legines brass barb 1/2 NPT x 5/8	$10.99	$10.99	Cast into the base, pointing down into the chip basket. +$7.59 shipping.
Ball valve	1/4 NPT	1	Beduan brass mini ball valve	$7.99	$7.99	
Manifold	1/4 NPT Y	1	FASPARTS brass Y block	$11.99	$11.99	One bullet says 1/2 in thread: check before ordering
Loc-Line nozzles	1/4 in kit	2	Loc-Line LP-40413 coolant hose kit	$15.15	$30.30	

Item	Spec	Qty	Listing	Price	Line	Notes
Push-to-connect fittings	8 mm x 1/4 NPT	1 (10)	Qjaiune push fittings	$7.49	$7.49	One at the pump outlet, so the return line unplugs and the sump slides out
Coolant concentrate	Semi-synthetic, 1 gal	1	Fusion Cool 2240	$52.95	$52.95	Mix 5-10 %
Way cover strip	0.1 mm 301 stainless	1	SPCCGG 301 strip 0.1 x 300 x 1000 mm	$54.10	$54.10	Gives all three strips. If it turns out soft, use 304 full-hard B09ZQQJ34Z for the 85 mm strips.
Way cover springs	Constant force, for printed rollers	1 (5-pack)	SUS301 constant-force springs, 5 pcs	$35.99	$35.99	One per narrow strip, two for the wide one
Spiral screw cover	20 mm ID, ~250-400 mm	1	MISONONEIK spiral steel bellows cover	?	?	Choose size 20-250-40-28 or 20-400-40-33; variants $47-116
Drip shield	510 x 57 mm	0	Cut from the ACM sheet offcut	$0	$0.00	Stainless was $57 (Speedy Metals 16 ga 12x24); ACM does the job

Priced subtotal: $573.92 + 3 unpriced: ACM sheet, Flood coolant pump, Spiral screw cover

Already owned

- HGR15-400 rails x2 + 4 HGH15CA blocks (X)
- SFU1204-400 kit with BK10/BF10, DSG12H housing, coupler (X)
- NEMA 23 bracket for BK10 (X)
- Open-loop NEMA 23 (Z)
- Extrusion: 10x 2040 x 400, 4x 2020 x 400, 4x 2020 x 200
- Bambu A1 mini, taps, drill press

Not included

- Cutting tools, a vise and clamps (workholding).
- The water hoses that come with the spindle kit (check the barb size before buying more).
- Wire, ferrules, fuses and breakers for the electrical box.

Check before ordering

1. **Couplers.** Both ballscrew kits ship with 6.35 mm-bore couplers; your motors have 8 mm shafts. Also check your owned open-loop motor's shaft and current.
2. **VFD on 120 V.** No real 2.2 kW drive takes 120 V and outputs 220 V; the step-up drives stop at 1.5 kW. The spindle runs at up to ~1.5 kW now and gets its full 2.2 kW when you switch to the kit's 220 V VFD on 240 V. Use a dedicated 20 A circuit.
3. **No cutting.** Rails, ballscrews and tube are used at the lengths they are sold in; the risers, plates and top plates are ordered cut to size. Only the M12 threaded rod is cut (hacksaw).
4. **Tubes and rebar.** Buy plain (black) steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale, degrease just before casting. Stock-length tube is +/-1/8 in on length, with an internal weld seam: file the seam where the spigots and printed sleeves go. 0.120 in wall (11 ga) works as well as 1/8 in.
5. **Imperial sizes.** The plates and risers are already inch sizes in the model. The 3/8 in top plates are 0.5 mm thinner than the modelled 10 mm, which doesn't matter.
6. **Low stock.** Several picks showed "only N left", including the closed-loop kits.